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On the dynamics of H_2 desorbing from a Pt(1 1 1) surface

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Abstract

A theoretical study of the dynamics of H_2 desorbing from a Pt(1 1 1) surface is reported. This study makes use of the ab initio potential energy surface (PES) of Olsen et al. [J. Chem. Phys. 116 (2002) 3841]. The topological study of the PES by the CHAIN method reveals unambiguously that the transition state (TS) is located at the top of a unique late barrier corresponding to a top site. The adsorbed H atoms are supposed to be in thermal equilibrium with the surface. In addition to that, classical trajectories show that recrossing of the TS is negligible. Accordingly, the nascent molecule H_2 is supposed to be still thermalized at the TS and batches of trajectories are run from the TS onto the products with their initial conditions canonically distributed. From the TS on, the total energy can be partitioned into three quantities: the vibrational energy, the rotational energy and the remaining energy, purely translational in the products only. It is found that on the average, the previous energy partitioning does not vary between TS and products. This finding is justified by the late barrier and the lightness of H_2. Last but not least, it is shown that transition state theory leads to a straightforward, but nevertheless accurate description of final state distributions.
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1. Introduction

Recombinative desorption and dissociative adsorption are two fundamental processes in gas–surface chemistry. This is why their dynamics has been widely studied both experimentally and theoretically during the last 50 years [1]. Since the pioneering work of van Willigen in 1968 [2], some of these works concern the recombinative desorption of hydrogen from metal surfaces. In the last 10 years, for instance, a lot of publications have been devoted to the dynamics of desorption of H_2 and D_2 from Cu(1 1 0) and Cu(1 1 1); the H_2/Cu system is in fact considered as an important model system for the understanding of activated gas–surface reactions. Without mentioning all these works, let us quote two recent reviews on

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this (and related) topic, respectively written by Hodgson [3] and Michelsen et al. [4].

In the present paper, we report a theoretical study of the dynamics of H_2 desorbing from a Pt(1 1 1) surface. Accurate quantum chemical calculations show that this desorption involves a late barrier along the reaction path [5,6]. The adsorbed H atoms are supposed to be in thermal equilibrium with the surface. In addition to that, classical trajectories show that recrossing of the transition state (TS), roughly localized at the barrier top, is negligible. Therefore, H_2 should be still thermalized at the TS. Classical trajectories (CT) are then carried out from the barrier top onto the products with their initial conditions canonically distributed. In order to account for the final vibrational quantization of H_2, we apply the gaussian weighting procedure recently introduced in gas-phase reaction dynamics [7–14]. From the transition state (TS) on, the total energy is partitioned into three quantities: the vibrational energy, the rotational energy and the remaining energy, purely translational in the products only. CT calculations show that this energy partitioning is unaltered beyond the barrier top. Such a finding is justified by the late nature of the barrier and the lightness of H_2. It is then shown that in such a case, transition state theory (TST) [15] can be used for a straightforward description of final state distributions.

The paper is organized as follows. In Section 2, we present the coordinate system, the classical function of Hamilton as well as the ab initio potential energy surface (PES) used in the work. CT calculations are presented in Section 3. TST expressions of state distributions are derived in Section 4. Comparison between CT and TST results is performed in Section 5. We conclude in Section 6.

2. The H_2/Pt(1 1 1) system

The dynamical state of H_2 is defined by the following set of canonical coordinates $(X, P_X, Y, P_Y, Z, P_Z, r, p_r, \theta, p_\theta, \phi, p_\phi)$. X and Y are the coordinates of the center of mass G of the molecule in a plane parallel to the surface, Z is the distance between G and the surface, r is the distance between the two atoms, θ is the angle between the molecule and the surface normal, and ϕ is the azimuthal angle (see Fig. 1). P_X, P_Y, P_Z, p_r, p_θ and p_ϕ are the conjugate momenta of X, Y, Z, r, θ and ϕ, respectively. In the framework of this coordinate system, the corresponding Hamiltonian is given by

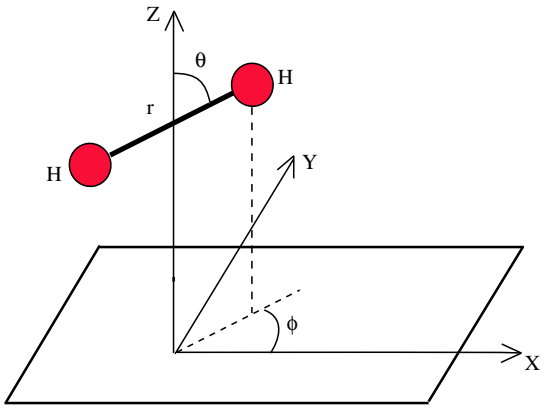


Fig. 1. Coordinate system for the H_2 + Pt(1 1 1) system.

$$H = \frac{P_X^2 + P_Y^2 + P_Z^2}{2M} + \frac{p_r^2}{2\mu} + \frac{J^2}{2I} + V \tag{2.1}$$

with

$$J^2 = p_\theta^2 + \frac{p_\phi^2}{\sin^2\theta} \tag{2.2}$$

and

$$I = \mu r^2. \tag{2.3}$$

M are μ are respectively the mass and the reduced mass of H_2. J is the rotational angular momentum of H_2 and I is its moment of inertia. The potential energy V is zero for Z tending to infinity and for $r = r_e$, where r_e is the equilibrium bond distance of the free H_2.

Surface degrees of freedom are not included in this study. As previously stated, classical trajectories are carried out from the barrier top onto the products. At the barrier top, the nascent molecule is already far from the surface so that the couplings between the H_2 degrees of freedom and the surface modes are extremely weak. In addition to that, the very light H_2 molecule recoils from the surface with a large velocity as compared with the average velocity of the much heavier Pt atoms. For

both reasons, we believe that not including the surface degrees of freedom in the present study is a very good approximation.

The potential energy V used in this work [5] was obtained from the corrugation-reducing procedure [16]. In order to characterize the topology of the corresponding PES, the potential calculation was implemented in the AMPAC 7.25 package [17]. The possible reaction paths between the adsorbed phase (two H atoms adsorbed on the surface) and the gas phase were then determined using the CHAIN method [18,19]. Briefly, this method consists in (a) defining a reactant and a product configuration, (b) generating a series of geometries forming a smooth continuous path between the two previous configurations, and (c) relaxing the geometries in directions perpendicular to the chain path. For activated processes, this procedure, detailed in Refs. [18,19] (see also Ref. [20]), leads to the saddle point corresponding to the highest energy along the relaxed path (other saddle points may exist along the relaxed path but their energy is lower).

Product configurations were defined by $Z = 6$ Å, $r = r_e$ and a random selection of X, Y, θ and ϕ. For the set of adsorbed configurations, r was kept at 2.77 Å, the maximum value available for the PES of Olsen et al. [5]. X, Y, Z, θ and ϕ were then randomly assigned within the restriction $Z < 3$ Å.

We ran a batch of 500 CHAIN calculations which led to the same saddle point. We therefore concluded that each trajectory corresponding to H_2 desorbing from a Pt(1 1 1) surface should go through the barrier corresponding to this saddle point.

This point is defined by $X^\ddagger = 0$, $Y^\ddagger = 0$, $Z^\ddagger = 2.24$ Å, $r^\ddagger = 0.78$ Å, $\theta^\ddagger = \frac{\pi}{2}$ and $\phi^\ddagger = \frac{\pi}{6}$. The center of mass of H_2 is therefore located over a top site (see Fig. 2). The energy difference between this saddle point and the H_2 molecule in the gas phase at its equilibrium geometry is $V^\ddagger = 57$ meV. This value is in good agreement with the barrier threshold of the dissociative adsorption of H_2 on Pt(1 1 1) found by Crespos et al. [6].

As can be seen in Fig. 3, the desorption barrier defined by this saddle point is located in the exit

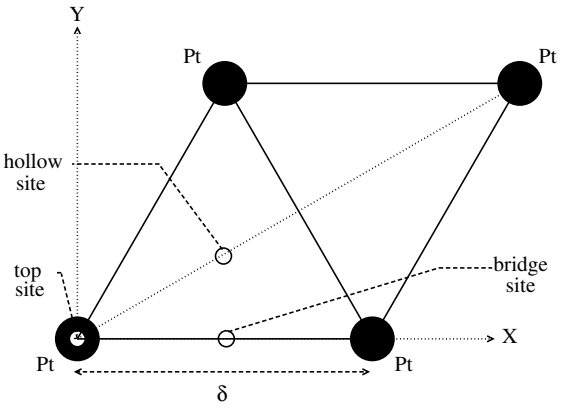


Fig. 2. High symmetry sites (top, bridge, hollow) on the Pt(1 1 1) surface. These sites are represented by open circles. δ corresponds to the smallest distance between two Pt atoms ($\delta = 2.77$ Å).

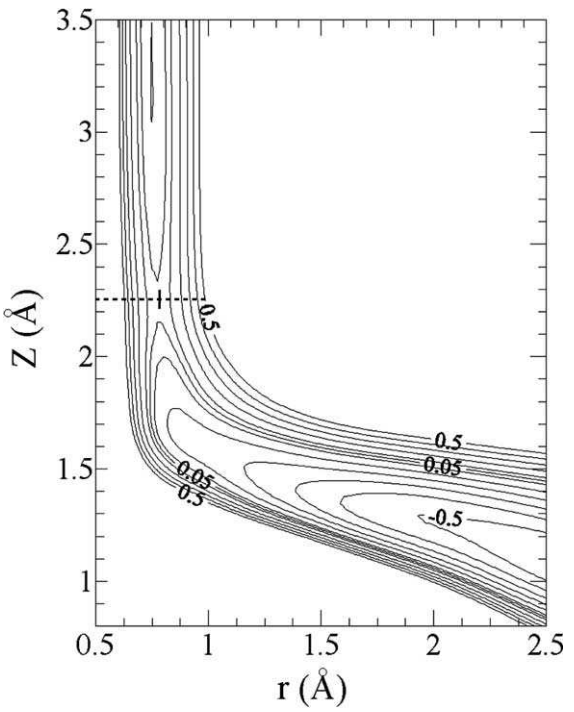


Fig. 3. Contour plot representation in the (r, Z) plane of the PES used in the calculation. The center of mass of the molecule is located over a top site with $\theta = \frac{\pi}{2}$ and $\phi = \frac{\pi}{6}$. The energy level spacing is 0.1 eV and the level 0.05 eV has been added. The dashed line in the figure represents the planar transition state defined by $Z = Z^\ddagger = 2.24$ Å.

channel. The desorption of H_2 from Pt(1 1 1) is thus characterized by a late barrier.

By calculating the eigenvectors of the hessian matrix at the saddle point, we obtained the direction of the reaction coordinate in the neighbourhood of this point. We found that an excellent approximation of the reaction coordinate is the Z coordinate. For the sake of convenience, we shall then define the dividing surface, i.e. the TS, by the simple equation $Z = Z^{\ddagger} = 2.24$ Å (see Fig. 3). In other words, the TS is supposed to be planar in our coordinate system.

3. Classical trajectory calculations

3.1. Methodology

Stricto sensus, our definition of the TS is only valid in the neighbourhood of the saddle point. As a consequence, some trajectories may cross the planar TS in the direction of the products and recross it back to the reagents without reaching the real TS. An analogous remark holds for non reactive trajectories coming from the products. In addition to that, it cannot be excluded that some trajectories may recross the real (and/or the planar) TS. However, we have checked that the percentage of trajectories of this kind is negligible.

While adsorbed on the Pt(1 1 1) surface, the two hydrogen atoms are supposed to be in thermal equilibrium, at temperature T_s, with the metal surface. At the TS, the nascent molecule can still be considered as being in thermal equilibrium with the metal surface since recrossing of the TS is negligible (this is a necessary condition—see, for instance, Ref. [21] and references therein). In the completely classical picture, the distribution of its phase space states is canonical and the desorption flux through the TS is given by [21]:

$$F^{\ddagger}(T_s) = \int \mathrm{d}\mathbf{q}\,\mathrm{d}\mathbf{p}\,\mathrm{d}P_Z \frac{P_Z}{M} \times \exp[-\beta H(\mathbf{q}, Z^{\ddagger}, \mathbf{p}, P_Z)]\Theta(P_Z) \qquad (3.1)$$

with $\mathbf{q} = (X, Y, r, \theta, \phi)$, $\mathbf{p} = (P_X, P_Y, p_r, p_\theta, p_\phi)$ and H defined in Eq. (2.1). $\beta = 1/(k_B T_s)$ where k_B is the Boltzmann constant. Θ is the Heaviside function. The initial conditions of the ith trajectory are therefore $\mathbf{q}_i$, $Z^{\ddagger}$, $\mathbf{p}_i$ and P_{Zi} (with $P_{Zi} \geqslant 0$) and its statistical weight is

$$W_i^{\ddagger} \propto P_{Zi} \exp[-\beta H(\mathbf{q}_i, Z^{\ddagger}, \mathbf{p}_i, P_{Zi})]. \qquad (3.2)$$

The total energy E of each trajectory ($E = H$) is allowed to vary between the value of the potential barrier $V^{\ddagger}$ and $10kT_s$, a value above which the weighting factor is negligible.

We shall not work, however, in the framework of the fully classical picture. Rather, we shall assume that the final vibration motion of H_2 is quantized. A convenient way to deal with this quantization is to run gaussian weighted trajectories [7–14] according to the following procedure. The Hamiltonian function can be written as (Eq. (2.1)):

$$H = H_V + H_R + H_T \qquad (3.3)$$

with

$$H_V = \frac{p_r^2}{2\mu} + V, \qquad (3.4)$$

$$H_R = \frac{J^2}{2I} \qquad (3.5)$$

and

$$H_T = \frac{P_X^2 + P_Y^2 + P_Z^2}{2M}. \qquad (3.6)$$

H_V, H_R and H_T are the vibrational, rotational and translational Hamiltonians in the separated products. The vibration action x of the final H_2 is thus defined by

$$x = \frac{H_V}{\hbar\omega} - \frac{1}{2} \qquad (3.7)$$

with ω the vibrational frequency of H_2.

The ith trajectory is then weighted by the following factor:

$$\delta_i = \sum_{v_i} \delta_{v_i}, \qquad (3.8)$$

where v_i is the vibrational quantum number ($v_i = 0, 1 \ldots$) and

$$\delta_{v_i} = \exp\left[\frac{(v_i - x_i)^2}{\epsilon^2}\right] \qquad (3.9)$$

is a gaussian function (x_i is the vibration action of the ith trajectory).

When the vibrational action is close to an integer value, the weighting factor is large. On the opposite, a vibrational action far from an integer value leads to a weighting factor close to zero.

The full-width-half-maximum ϵ of the gaussian functions was chosen small enough to quantize the vibration but sufficiently large to get a satisfying statistics. ϵ was kept at 0.1.

The overall weighting factor W_i^f of the ith trajectory in the products is therefore:

$$W_i^f = \delta_i W_i^{\ddagger}. \tag{3.10}$$

Gaussian weighting in CT calculations has been recently applied to several gas-phase chemical reactions and good agreement has been found with quantum scattering calculations when tunneling is negligible [9–14]. This procedure has been recently justified in the framework of classical S-matrix [8], i.e. the semiclassical theory of molecular collisions proposed by Miller and Marcus in the early seventies (see Refs. [5–7] in Ref. [8]).

In the following, we shall also consider the following partition of the total energy:

$$H = H_{\mathrm{V}}^{\ddagger} + H_{\mathrm{R}} + H_{\mathrm{NRV}}^{\ddagger} \tag{3.11}$$

with

$$H_{\mathrm{V}}^{\ddagger} = \frac{p_r^2}{2\mu} + V_r^{\ddagger}(r), \tag{3.12}$$

$$V_r^{\ddagger}(r) = \frac{1}{2}k_r(r - r^{\ddagger})^2, \tag{3.13}$$

k_r being the stretching force constant at the saddle point and

$$\begin{aligned} H_{\mathrm{NRV}}^{\ddagger} &= H - H_{\mathrm{V}}^{\ddagger} - H_{\mathrm{R}} \\ &= \frac{P_X^2 + P_Y^2 + P_Z^2}{2M} + V - V_r^{\ddagger}(r). \end{aligned} \tag{3.14}$$

$H_{\mathrm{V}}^{\ddagger}$ is the vibrational Hamiltonian, H_{R} is given by Eq. (3.5) and $H_{\mathrm{NRV}}^{\ddagger}$ is the remaining non rovibrational part of the total Hamiltonian. The vibration action at the TS is given by

$$x^{\ddagger} = \frac{H_{\mathrm{V}}^{\ddagger}}{\hbar\omega^{\ddagger}} - \frac{1}{2}, \tag{3.15}$$

$\omega^{\ddagger}$ being the vibrational frequency at the saddle point.

3.2. Dynamical results

The surface temperature T_s was kept at 850 K. The CT calculations involved 2×10^5 reactive trajectories. The final distributions of the rotational, translational and vibrational energies as well as the distribution of the total energy (Eqs. (3.3)–(3.6)) are represented in Fig. 4. Because of the gaussian-weighting procedure, the final vibrational energy distribution is a gaussian function centered at the H_2 zero point energy (0.27 eV). The mean value of the vibrational action is -0.01 at the TS and -0.03 in the products, a value very close to zero. The vibration motion of H_2 is thus adiabatic beyond the TS. Moreover, the level spacing does not vary significantly in this region. The vibrational energy is thus roughly a constant of motion in the exit-channel.

In Fig. 5, the value E_{T} of the translational energy H_{T} in the final products (Eq. (3.6)) is plotted against the value $E_{\mathrm{NRV}}^{\ddagger}$ of the non rovibrational energy $H_{\mathrm{NRV}}^{\ddagger}$ at the TS (Eq. (3.14)). We note that on the average:

$$E_{\mathrm{T}} = E_{\mathrm{NRV}}^{\ddagger}. \tag{3.16}$$

Therefore, the potential term $V - V_r^{\ddagger}(r)$ of the non rovibrational hamiltonian (see Eq. (3.14)) is

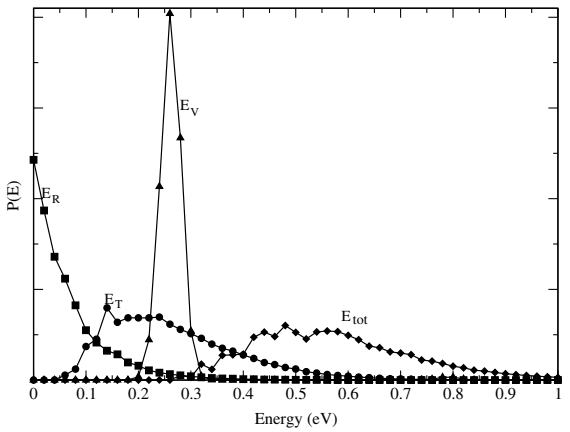


Fig. 4. Final energy distributions of the desorbed H_2; squares: rotational energy distribution; circles: translational one; triangles: vibrational one; diamonds: total energy distribution.

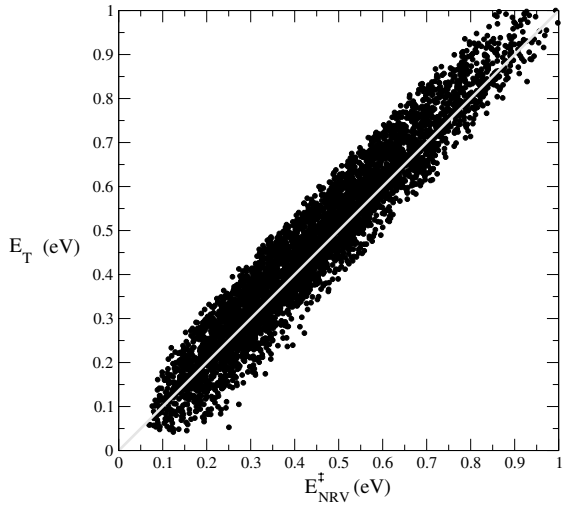


Fig. 5. Translational energy in the product region E_T plotted as a function of the energy $E^{\ddagger}_{NRV}$ (dots). The grey solid line corresponds to $E_T = E^{\ddagger}_{NRV}$.

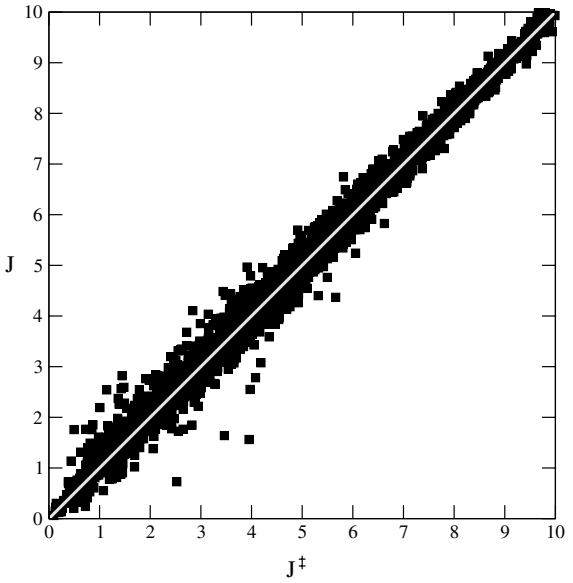


Fig. 6. Angular momentum J in the products plotted as a function of the angular momentum $J^{\ddagger}$ at the transition state (squares). The grey solid line corresponds to $J = J^{\ddagger}$.

released into translational energy in the products (see Eq. (3.6)). From now on, we will not differentiate $E^{\ddagger}_{NRV}$ and E_T which will be both denoted by E_T. The final translational energy distribution $P(E_T)$ is then the same as the TS one:

$$P(E_T) = P^{\ddagger}(E_T). \tag{3.17}$$

The mean value of the final rotational energy, depicted in Fig. 4, is 71 meV, i.e. nearly kT_s (74 meV at $T_s = 850$ K). The value of the angular momentum J in the products is given in Fig. 6 as a function of the angular momentum $J^{\ddagger}$ at the TS. We can see that on the average the rotation motion is unaltered beyond the TS:

$$J = J^{\ddagger}. \tag{3.18}$$

The final rotational state distribution $P(J)$ is thus the same as the TS one:

$$P(J) = P^{\ddagger}(J). \tag{3.19}$$

The fact that the energy partitioning is conserved on the way from the TS to the products can be explained as follows: the desorption barrier being late, the torque on H_2 for θ values different from $\frac{\pi}{2}$ is rather weak. In addition to that, H_2 is very light. Accordingly, the time spent by the molecule in the interaction region is expected to be very short. In other words, there is no time for the energy transfer between rotation and translation motions to be efficient. In the next section, we shall use this finding to show that TST leads to a straightforward description of final state distributions of the desorbed H_2 molecule.

4. Transition state theory description of state distributions

4.1. Assumptions

(1) We have assumed that the adsorbed H atoms are in thermal equilibrium with the surface. Moreover, we have seen that recrossing of the TS is negligible. These two statements, which are actually the basic assumptions of TST, make canonical the distribution of the phase space states of H_2 at the TS (see Ref. [21] and references therein).

(2) As the desorption barrier is late, we assume that from the TS onto the products, the interaction potential does not depend on the X, Y and ϕ coordinates. We thus neglect the surface corrugation.

Moreover, at the TS, we have found that a reasonable approximation of the potential energy is given by

$$V = V^{\ddagger} + V_r^{\ddagger}(r) + V_\theta^{\ddagger}(\theta) \tag{4.1}$$

with $V_r^{\ddagger}(r)$ given by Eq. (3.13) and

$$V_\theta^{\ddagger}(\theta) = \alpha\cos^2\theta. \tag{4.2}$$

$V^{\ddagger}$ is the barrier threshold previously introduced and $V_\theta^{\ddagger}(\theta)$ is the term associated with the bending motion of H_2 at the TS. The bending motion is thus supposed to be limited to a plane orthogonal to the surface. k_r and α, the stretching and bending force constants at the TS, are respectively equal to 31.5 eV Å^{-2} and 0.162 eV. Eq. (4.2) respects the π periodicity and reaches a minimum when $\theta = \theta^{\ddagger} = \frac{\pi}{2}$. Within this definition of the interaction potential at the TS, the Hamiltonian $H_{\mathrm{NRV}}^{\ddagger}$ (Eq. (3.14)) can be written as

$$H_{\mathrm{NRV}}^{\ddagger} = \frac{P_X^2 + P_Y^2 + P_Z^2}{2M} + V_{\mathrm{T}}, \tag{4.3}$$

where

$$V_{\mathrm{T}} = V - V_r^{\ddagger}(r) = V^{\ddagger} + V_\theta^{\ddagger}(\theta). \tag{4.4}$$

(3) At the TS, all the orientations of the H_2 molecule are possible. Nevertheless, within the canonical distribution framework, configurations corresponding to a molecule parallel to the surface are more likely. We thus assume that the angle θ is fluctuating around the value $\theta^{\ddagger} = \frac{\pi}{2}$. We shall thus replace in Eq. (2.2), the expression $\sin^2\theta$ by its average value $\langle\sin^2\theta\rangle$ which is a constant slightly smaller than 1. At the TS, we have thus:

$$J^2 \simeq p_\theta^2 + \frac{p_\phi^2}{\langle\sin^2\theta\rangle}. \tag{4.5}$$

4.2. Translational energy distribution

According to TST (Ref. [15] and references therein, Ref. [22]), the distribution of the non rovibrational energy E_{T} at the TS is defined by

$$P^{\ddagger}(E_{\mathrm{T}}) = \frac{F^{\ddagger}(E_{\mathrm{T}}, T_{\mathrm{s}})}{F^{\ddagger}(T_{\mathrm{s}})}, \tag{4.6}$$

where $F^{\ddagger}(T_{\mathrm{s}})$ is the canonical flux of trajectories through the TS. Unlike Eq. (3.1), $F^{\ddagger}(T_{\mathrm{s}})$ is only due here to trajectories for which the vibrational action of H_2 has an integer value from the TS on. $F^{\ddagger}(E_{\mathrm{T}}, T_{\mathrm{s}})$ is the contribution of $F^{\ddagger}(T_{\mathrm{s}})$ due to trajectories crossing the TS with a non rovibrational energy E_{T}. We recall that the non rovibrational energy at the TS becomes the translational energy in the products. Accordingly, the above density is also the final translational energy distribution $P(E_{\mathrm{T}})$ (see Eq. (3.17)).

$P(E_{\mathrm{T}})$ is thus directly proportional to $F^{\ddagger}(E_{\mathrm{T}}, T_{\mathrm{s}})$ given by

$$F^{\ddagger}(E_{\mathrm{T}}, T_{\mathrm{s}}) = \sum_{v=0}^{\infty} F^{\ddagger}(E_{\mathrm{T}}, v, T_{\mathrm{s}}) \tag{4.7}$$

with

$$F^{\ddagger}(E_{\mathrm{T}}, v, T_{\mathrm{s}}) = \int \mathrm{d}\mathbf{\Gamma}\exp(-\beta H) \times \frac{P_Z}{M}\Theta(P_Z)\delta_1\delta_2\delta_3, \tag{4.8}$$

$$\mathrm{d}\mathbf{\Gamma} = \mathrm{d}X\,\mathrm{d}P_X\,\mathrm{d}Y\,\mathrm{d}P_Y\,\mathrm{d}Z\,\mathrm{d}P_Z\,\mathrm{d}r\,\mathrm{d}p_r\,\mathrm{d}\theta\,\mathrm{d}p_\theta\,\mathrm{d}\phi\,\mathrm{d}p_\phi, \tag{4.9}$$

$$\delta_1 = \delta(Z - Z^{\ddagger}), \tag{4.10}$$

$$\delta_2 = \delta\left(v + \frac{1}{2} - \frac{H_{\mathrm{V}}^{\ddagger}}{\hbar\omega}\right) \tag{4.11}$$

and

$$\delta_3 = \delta(E_{\mathrm{T}} - H_{\mathrm{NRV}}^{\ddagger}). \tag{4.12}$$

β is equal to $(kT_{\mathrm{s}})^{-1}$ and H is defined by Eq. (2.1). δ_1 implies the flux calculation through the TS, δ_2 limits the integration to the states corresponding to the value v of the vibrational action, and δ_3 limits the integration to the states corresponding to the value E_{T} of the non rovibrational energy.

It is shown in the appendix that:

$$F^{\ddagger}(E_{\mathrm{T}}, v, T_{\mathrm{s}}) \propto \exp[-\beta(E_{\mathrm{T}} + \hbar\omega v)] \times \int_0^{\pi} \mathrm{d}\theta\, f(\theta, E_{\mathrm{T}})\Theta[f(\theta, E_{\mathrm{T}})] \tag{4.13}$$

with

$$f(\theta, E_T) = E_T - V^{\ddagger} - \alpha\cos^2\theta. \quad (4.14)$$

We finally arrive at:

$$P(E_T) \propto \exp(-\beta E_T) \times \int_0^{\pi} d\theta\, f(\theta, E_T)\Theta[f(\theta, E_T)], \quad (4.15)$$

which can be trivially evaluated numerically.

4.3. Rotational state distribution

The distribution of the angular momentum J at the TS is defined by

$$P(J) = \frac{F^{\ddagger}(J, T_s)}{F^{\ddagger}(T_s)}, \quad (4.16)$$

where $F^{\ddagger}(J, T_s)$ is the canonical flux of trajectories crossing the TS with a given angular momentum J. $P(J)$ is thus directly proportional to $F^{\ddagger}(J, T_s)$, given by the following expression:

$$F^{\ddagger}(J, T_s) = \sum_{v=0}^{\infty} F^{\ddagger}(J, v, T_s) \quad (4.17)$$

with

$$F^{\ddagger}(J, v, T_s) = \int d\Gamma \exp(-\beta H) \times \frac{P_Z}{M}\Theta(P_Z)\delta_1\delta_2\delta_4 \quad (4.18)$$

and

$$\delta_4 = \delta\left[J - (2IH_R^{\ddagger})^{\frac{1}{2}}\right] \quad (4.19)$$

($d\Gamma$, δ_1 and δ_2 are defined as previously).

Proceeding as in the appendix, it is found that:

$$F^{\ddagger}(J, v, T_s) \propto J \exp\left[-\beta\left(\frac{J^2}{2I} + \hbar\omega v\right)\right]. \quad (4.20)$$

We finally arrive at:

$$F^{\ddagger}(J, T_s) \propto J \exp\left[-\frac{\beta J^2}{2I}\right]. \quad (4.21)$$

The normalised expression of the rotational state distribution is thus given by:

$$P(J) = \frac{\beta}{I} J \exp\left[-\frac{\beta J^2}{2I}\right]. \quad (4.22)$$

5. Comparison between CT and TST calculations

The translational energy and rotational state distributions found from CT and TST calculations are represented in Figs. 7 and 8 for a temperature T_s of 850 K. The agreement between both approaches is very satisfying, as expected from the

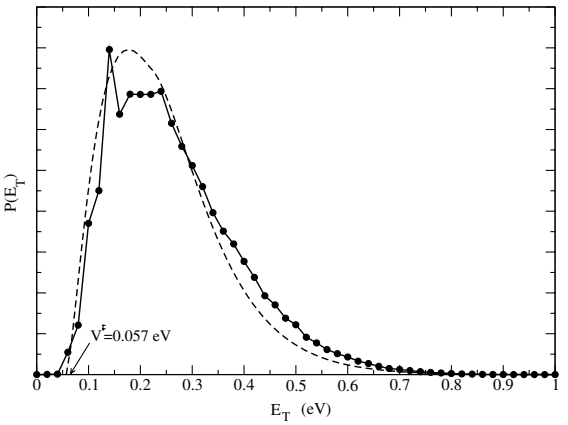


Fig. 7. Translational energy distribution obtained by CT calculations in the product region (filled circles connected by segments). The TST translational energy distribution is represented by a dashed line.

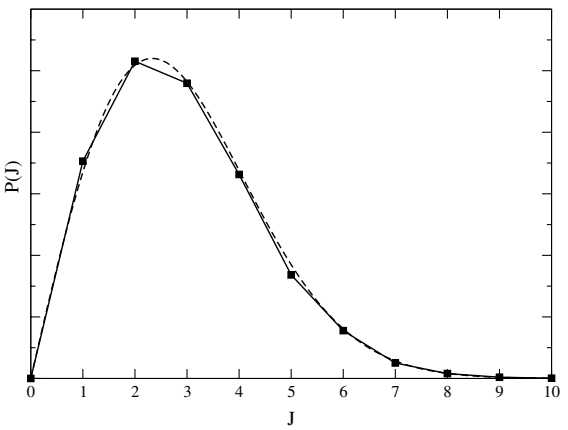


Fig. 8. Rotational state distribution obtained by CT calculations in the products region (filled squares; the solid line is a guide for the eyes). The dashed line represents the TST rotational state distribution.

invariance of the energy partitioning in the exit-channel discussed in Section 3.2.

6. Conclusion

A theoretical study of the dynamics of H_2 desorbing from a Pt(1 1 1) surface has been reported. The main conclusions of the study are as follows:

(1) For the ab initio PES of Olsen et al. [5], the CHAIN method of Liotard and coworkers [18,19] shows unambiguously that the TS is located at the top of a unique barrier along the reaction path. Moreover, the barrier is late so that the TS is well defined by the value $Z^{\ddagger} = 2.24$ Å between the Pt(1 1 1) surface and the center of mass of H_2.

(2) The adsorbed H atoms are supposed to be in thermal equilibrium with the surface. Since recrossing of the TS is found to be negligible, the nascent H_2 molecule is expected to be still thermalized at the TS. Batches of trajectories have thus been run from the TS onto the products with their initial conditions canonically distributed. The gaussian weighting procedure recently introduced in gas-phase reaction dynamics has been applied in order to account for the final vibrational quantization of H_2 [7–14].

From the TS on, the total energy is partitioned into three quantities: the vibrational energy, the rotational energy and the non rovibrational energy, purely translational in the products only. We have found that on the average, this energy partitioning does not vary between TS and products. The reasons are as follows: (a) the barrier is late; the vibration motion is adiabatic and the angular torque that H_2 experiences when going down to the barrier is rather small. (b) H_2 is a very light molecule which does not spend enough time on the product side of the barrier for its rotation motion to be altered by the small torque. So is the non rovibrational energy which becomes purely translational in the final products.

(3) For such a system, TST can be used for a straightforward and accurate description of final state distributions.

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Appendix

Integrating Eq. (4.8) with respect to the Z variable results in

$$F^{\ddagger}(E_T, v, T_s) = \int d\mathbf{\Gamma}' \exp(-\beta H) \times \frac{P_Z}{M} \Theta(P_Z)\delta_2\delta_3 \tag{A.1}$$

with

$$d\mathbf{\Gamma}' = dX\, dP_X\, dY\, dP_Y\, dP_Z\, dr\, dp_r\, d\theta\, dp_\theta\, d\phi\, dp_\phi. \tag{A.2}$$

H is given in Eq. (3.11) and δ_2 and δ_3 are defined in Section 4.2.

From Eqs. (3.12)–(3.14), (4.2)–(4.4), we can notice that the X, Y and ϕ variables do not appear in H. We can thus trivially integrate with respect to these variables, then arriving at

$$F^{\ddagger}(E_T, v, T_s) \propto \int dP_X\, dP_Y\, dP_Z\, dr\, dp_r\, d\theta\, dp_\theta\, dp_\phi \times \exp(-\beta H)\frac{P_Z}{M}\Theta(P_Z)\delta_2\delta_3. \tag{A.3}$$

From Eqs. (3.11), (3.5) and (4.5), we can integrate with respect to p_θ and p_ϕ, leading to

$$F^{\ddagger}(E_T, v, T_s) \propto \int dP_X\, dP_Y\, dP_Z\, dr\, dp_r\, d\theta \times \exp[-\beta(H^{\ddagger}_{NRV} + H^{\ddagger}_{V})]\frac{P_Z}{M}\Theta(P_Z)\delta_2\delta_3. \tag{A.4}$$

Setting

$$\rho\cos\psi = \sqrt{\frac{p_r^2}{\mu\hbar\omega}} \tag{A.5a}$$

and

$$\rho\sin\psi = \sqrt{\frac{k_r(r - r^{\ddagger})^2}{\hbar\omega}}, \tag{A.5b}$$

Eq. (A.4) becomes:

$$F^{\ddagger}(E_T, v, T_s) \propto \int dP_X\, dP_Y\, dP_Z \rho\, d\rho\, d\psi\, d\theta \times \exp[-\beta(H^{\ddagger}_{NRV} + H^{\ddagger}_{V})] \frac{P_Z}{M} \Theta(P_Z)\delta_2\delta_3 \tag{A.6}$$

with

$$H^{\ddagger}_{V} = \frac{\hbar\omega}{2}\rho^2. \tag{A.7}$$

After integration with respect to ρ and ψ, we have:

$$F^{\ddagger}(E_T, v, T_s) \propto \int dP_X\, dP_Y\, dP_Z\, d\theta \times \exp[-\beta(H^{\ddagger}_{NRV} + \hbar\omega v)] \frac{P_Z}{M} \Theta(P_Z)\delta_3. \tag{A.8}$$

Since

$$dP_Z\left(\frac{P_Z}{M}\right) = d\left(\frac{P_Z^2}{2M}\right) = dx \tag{A.9}$$

and

$$H^{\ddagger}_{NRV} = \frac{P_X^2 + P_Y^2}{2M} + V_T + x, \tag{A.10}$$

integration with respect to x leads to

$$F^{\ddagger}(E_T, v, T_s) \propto \exp[-\beta(E_T + \hbar\omega v)] \int dP_X\, dP_Y\, d\theta \times \Theta\left[E_T - \left(\frac{P_X^2 + P_Y^2}{2M} + V_T\right)\right] \tag{A.11}$$

with V_T defined in Eq. (4.4).

The final integration with respect to P_X and P_Y leads to Eqs. (4.13) and (4.14).

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